

BINGYI LIU (EDISON)

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Research Interests: Human-AI Interaction | Spatial & Egocentric AI | XR & Wearable Computing

Education

Purdue University

Expected December 2026

B.S. in Computer Science, Machine Intelligence Track | GPA: 3.91/4.00

West Lafayette, Indiana

Awards & Honors: Purdue SURF (\$6,500); Dean's List; Semester Honors

Teaching & Leadership: Teaching Assistant, CS 18000; Executive Director, Purdue Computer Pioneers Union

Selected Publications & Manuscripts

- **Situated Human-Agent Co-Reasoning in AR.** *Submitted to ACM CHI 2027.*
- **In-Situ Authoring for Embodied Agents in AR.** *Submitted to ACM CHI 2027.*
- Yiwei Li, Zhitong Guo, **Bingyi Liu**, and Yan Li. "Planner-Validated Genetic Learning of Lifted Goal Agendas for Goal-Dependent PDDL Planning." *Accepted at CSAI 2026.*

Research Experience

Purdue University C Design Lab

October 2025 – Present

Undergraduate Research Assistant | Advised by Prof. Karthik Ramani

West Lafayette, Indiana

In-Situ Authoring for Embodied Agents in AR

Oct. 2025 – Mar. 2026

- Introduced a hierarchical **experience-episode-unit interaction graph** for representing spatial context, agent behaviors, and conditional transitions, paired with a dual-view AR interface for in-situ authoring and world-in-miniature task overview.
- Developed a **multimodal compiler/runtime** that resolves speech, gaze, pointing, and user pose into spatially grounded scene referents, compiles them into graph-structured interaction units, and executes dialogue, navigation, deictic, and conditional behaviors on Meta Quest.
- Evaluated the end-to-end system in a two-session study with **12 participants**, achieving a **79.8 ± 10.8 SUS** score; three core generation/runtime modules reached **97%–100% valid-output rates**.

Spatially Grounded Human-Agent Reasoning in AR

Jun. 2026 – Present

- Architected a **stateful, tool-using reasoning agent** for situated AR tasks, separating generative reasoning from deterministic execution; the agent routes wearer goals and first-person scene context through a TaskFrame compiler, Visual Observer, and Public Reasoner to select the next perception, spatial-query or human-solicitation.
- Built the agent's **persistent spatial state and memory** with versioned StateDeltas, bounded reasoning context, SceneGraph entities, and task-selected keyframes, integrating VLM semantics, YOLOE-Seg, and Quest depth/pose for first-person 2D/3D scene understanding while preventing stale or duplicate actions from committing.
- Designed a **human-in-the-loop co-reasoning and revision mechanism** that lets the agent request missing situated knowledge or user judgment, incorporates user corrections into provenance-aware reasoning state, and propagates revisions only through dependent branches while preserving unaffected state and interaction history.

Purdue University Elmore Family School of ECE

February 2025 – May 2026

Research Assistant | Advised by Prof. Yung-Hsiang Lu

West Lafayette, Indiana

Compositional and Hardware-Aware Evaluation of Text-to-Image Models

- Designed a compositional evaluation benchmark for seven text-to-image models spanning diffusion, GAN, and autoregressive families across object, count, attribute, spatial-relation, and style factors; **fine-tuned CLIP** on the project dataset as a task-specific semantic-alignment classifier, improving held-out classification accuracy from **81.6% to 90.8%** and macro-F1 from **0.79 to 0.89** over the off-the-shelf encoder.
- Engineered a **reproducible edge-inference stack** for seven generative models on NVIDIA Jetson Orin, standardizing warm-up, CUDA-synchronized timing, mixed precision, and GPU-memory profiling across 5 evaluation metrics.

Accuracy-Oriented 3D Piano Hand Motion Generation | *Project Lead*

- Reformulated MIDI-conditioned bimanual motion generation around **time-aligned 3D key-contact targets**, mapping pitch, onset, and offset directly to fingertip-key constraints and training a music-conditioned flow-matching model with contact, pose-prior, smoothness, and motion-preservation objectives; improved **key-hit F1 from 0.71 to 0.82** and reduced fingertip-key contact error by **27.4%** over an unconstrained baseline.
- Developed **event-local refinement** around note onsets and active key intervals; on hard passages with large leaps, dense chords, and rapid transitions, refinement reduced contact-timing error from **92 ms to 64 ms** and wrong-key events by **24.8%**, while changing MPJPE by less than **3%**.

- Designed and evaluated an end-to-end **computer-vision pipeline** for real-time fruit recognition, benchmarking **five CNN backbones** while tuning input resolution and preprocessing—including Sobel edge filtering, resizing, and RGB normalization; achieved **95% accuracy** with under **200 ms** processing time.
- Built a **fault-tolerant Android data pipeline** with Kotlin Coroutines, WorkManager, Retrofit, and OkHttp, supporting asynchronous capture/upload, retry-aware execution, and over **99% successful uploads**.

Industry Experience

Action Intelligence

Egocentric AI Algorithm Intern | Supervised by Dr. Jingkang Yang

August 2026 – Present

Shanghai, China

- Investigating **long-horizon egocentric video understanding** for intelligent eyewear, organizing visual-audio observations into event-centric episodic memory that preserves entities, interactions, state changes, and temporal dependencies across continuous first-person streams.
- Developing a tool-augmented long-video reasoning pipeline with hierarchical retrieval and retrieve-inspect-verify planning, enabling grounded event recall, temporal localization, and personalized question answering over hour- to day-scale experience.
- Contributing to egocentric data curation, post-training, and benchmark design for event retrieval, and multimodal QA, while profiling the accuracy-latency trade-offs required for deployable wearable assistants.

Eyedaptic

AI Software Engineer Intern

Summer 2025 & Summer 2026

Laguna Hills, California

Eva YouTube Video Assistant Agent for AR Glasses

May. 2026 – Jul. 2026

- Architected a **stateful, tool-using voice agent** for Eyedaptic's Android AR-glasses platform, orchestrating **20+ typed tools** through Gemini Live function calling for search, multi-turn refinement, selection, confirm-before-play, history resume, and playback control.
- Built the agent's **retrieval and personalization layer** with the YouTube Data API, multi-signal reranking, channel-aware context, and SQLite behavioral memory; improved mean top-1 ranking score from **0.182 to 0.604** and outperformed the baseline in **10/10 evaluation scenarios**.

Multimodal AR-Glasses Assistant

May. 2025 – Jul. 2025

- Engineered a **real-time multimodal agent runtime** for low-vision AR-glasses users, coupling first-person visual/OCR perception with concurrent response generation, interruptible streaming PCM, and **TaskToken-based cancellation**; reduced time-to-first-speech by **2–3 s** and long-form latency by up to **80%**.

Tongcheng Travel

Software Engineer Intern

June 2023 – August 2023

Suzhou, China

- Implemented concurrent order creation, inventory deduction, and idempotency in **Spring Boot**, combining Redis Lua atomic updates, MySQL optimistic locking, and request keys to prevent duplicate orders and overselling.
- Designed cache-aside and pre-warming for event/seat queries, maintaining approximately **110 ms p95 latency** under peak-load tests; supported Kubernetes canary releases and rollback drills with approximately **3 min MTTR**.

Selected Projects

SEEAgent: Egocentric Perception-to-Action Agent for Smart Glasses

- Building a **stateful, tool-using egocentric agent** that turns first-person video and spoken intent into executable digital actions, combining cross-frame memory, scene understanding, checkpointed tool planning, and result verification to support multi-step tasks that can be interrupted, revised, and resumed.

BeatDopamine: Adaptive Agent for Attention and Self-Regulation

- Designing an **adaptive behavioral agent** that learns how a user responds to distraction, effort, delay, and interruption, then adjusts commitment size, task difficulty, and intervention timing; temporary scaffolding gradually fades as sustained attention and self-directed control recover.

Technical Skills

Programming & ML: Python, C/C++, C#, Java/Kotlin, Swift, Go; PyTorch, Transformers, OpenCV, CUDA, TensorRT

AI & Agents: VLMs, RAG/memory, multimodal systems, tool/function calling, MCP, LangGraph, agent evaluation

Platform: Linux/SLURM, Docker, Kubernetes, FastAPI, Redis, Git; Unity, Android/iOS, NVIDIA Jetson, AWS/GCP/Azure

AI-native builder; I work with Codex and Claude Code as day-to-day collaborators.